4.1 $SST=\sum(y_i-\bar{y})^2=\sum y_i^2-N\bar{y}^2$

a. $N=20$, $\sum y_i^2=7825.94$, $\bar{y}=19.21$, $SSR=375.47$, 求 R^2

$$SST=\sum_{i=1}^{20}(y_i-\bar{y})^2=\sum_{i=1}^{20}(y_i^2-2y_i\bar{y}+\bar{y}^2)=\sum_{i=1}^{20}y_i^2-2\bar{y}\sum_{i=1}^{20}y_i+\sum_{i=1}^{20}\bar{y}^2$$

$$=7825.94-2\times19.21\times20\times19.21+20\times19.21^2=445.458$$

故所求 $R^2=\dfrac{SSR}{SST}=\dfrac{375.47}{445.458}=0.8428853$ #

b. $R^2=0.7911$, $SST=725.94$, $N=20$, 求 $\hat{\sigma}^2$

$$R^2=\frac{SSR}{SST}=1-\frac{SSE}{SST}=0.7911\Rightarrow SSE=(0.7911-1)\times(-725.94)$$
$$=151.648866$$

又 $\hat{\sigma}^2=\dfrac{SSE}{N-2}=\dfrac{151.648866}{18}=8.424937$ #

c. $\sum(y_i-\bar{y})^2=SST=631.63$, $\sum\hat{e}_i^2=SSE=182.85$, 求 R^2

$$R^2=\frac{SSR}{SST}=1-\frac{SSE}{SST}=1-\frac{182.85}{631.63}=0.7105109$$ #

4-3

$\bar{x}=1$, $\bar{y}=2$

$x_i = 3, 2, 1, -1, 0$. $y_i = 4, 2, 3, 1, 0$ $(N=5)$

fitted regression line: $\hat{y_i} = 1.2 + 0.8x_i$

$SSE = \sum_{i=1}^{5} \hat{e_i}^2 = 3.6$, $\sum_{i=1}^{5}(x_i-\bar{x})^2 = 10$, $\sum_{i=1}^{5}(y_i-\bar{y})^2 = 10$

a. when $x_0=4$, the predicted value $\hat{y_0} = 1.2 + 0.8 \times 4 = 4.4$ #

b. $\hat{var}(f) = \hat{\sigma}^2\left(1 + \frac{1}{N} + \frac{(x_0-\bar{x})^2}{\sum(x_i-\bar{x})^2}\right)$, where $\hat{\sigma}^2 = \frac{SSE}{3} = 1.2$

$\Rightarrow \hat{var}(f) = 1.2 \times \left(1 + \frac{1}{5} + \frac{3^2}{10}\right) = 2.52$

$\therefore se(f) = \sqrt{\hat{var}(f)} = \sqrt{2.52} = 1.58745 \doteqdot 1.5875$ #

c. given $x_0=4$, y_0 的 95% prediction interval $(\alpha = 0.05)$

$= \hat{y_0} \pm t_c \, se(f)$, where $t_c = t_{df=3,\,0.025} = 3.182$

$= 4.4 \pm 3.182 \times 1.5875$

$\doteqdot [-0.6520, 9.4520]$ #

d. given $x_0 = 4$ 的 99% prediction interval ($\alpha = 0.01$)

$= \hat{y}_0 \pm t_c\, se(f)$, where $t_c = t_{df=3,\,0.005} = 5.841$

$= 4.4 \pm 5.841 \times 1.5875$

$\doteqdot [-4.8722,\ 13.6722]$ #

e. given $x = \bar{x} = 1$ 的 95% prediction interval ($\alpha = 0.05$)

$\hat{y} = 1.2 + 0.8 \times 1 = 2 = \bar{y}$

$\therefore PI = \hat{y} \pm t_c\, se(f)$, where $t_c = 3.182$

$\left(\widehat{var}(f) = 1.2\left(1 + \frac{1}{5} + 0\right) = 1.44 \Rightarrow se(f) = 1.2\right)$

$= 2 \pm 3.182 \times 1.2 \doteqdot [-1.8189,\ 5.8189]$

可以看到當 $x = \bar{x} = 1$ 的 prediction interval 的寬度會比 $x = x_0 = 4$ 的 PI 來的更窄 #